Global Economic Shifts

ECON 220

Import Tariffs

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Questions to Consider

1. Why do governments sometimes apply a tariff on imported goods?
2. Why does the World Trade Organization try to reduce the use of tariffs?

Introduction (part 1)

- During the U.S. presidential campaign of 2016, the Republican candidate Donald Trump said that he would apply a 45% tariff against imports from China.
- President Trump did impose significant import tariffs on China in 2018–19.
- The import tariff is an example of a trade policy, a government action meant to influence the amount of international trade.
- Import tariffs are sometimes used for political and not economic reasons. This is called the political economy of tariffs.
- Because the gains from trade are unevenly spread, industries and labor unions often feel that the government should do something to help limit their losses (or maximize their gains) from international trade.
- That “something” is trade policy, which includes the use of import tariffs (taxes on imports), import quotas (quantity limits on imports), and export subsidies.

Introduction (part 2)

- In this lecture, we begin our investigation of trade policies by focusing on the effects of tariffs and quotas in a perfectly competitive industry.
- We explain the reasons why countries apply tariffs and the effects of these tariffs on the producers and consumers in the importing and exporting countries.

1) A Brief History of the World Trade Organization (part 1)

- After World War II, the Allied countries met to discuss issues such as high trade barriers and unstable exchange rates.
- In 1947 the General Agreement on Tariffs and Trade (GATT) was established to reduce barriers to trade between nations.
- Some of the GATT's main provisions are as follows:
  1. A nation must extend the same tariffs to all trading partners that are GATT members.
  2. Tariffs may be imposed in response to unfair trade practices such as dumping.

     "Dumping" is defined as the sale of export goods at a price less than that charged at home, or at a price less than costs of production and shipping.

1) A Brief History of the World Trade Organization (part 2)

Some of the GATT’s main provisions are as follows:

3. Countries should not limit the quantity of goods and services that they import.
4. Countries should declare export subsidies provided to particular firms, sectors, or industries.
5. Countries can temporarily raise tariffs for certain products.

1) A Brief History of the World Trade Organization (part 3)

Some of the GATT's main provisions are as follows:

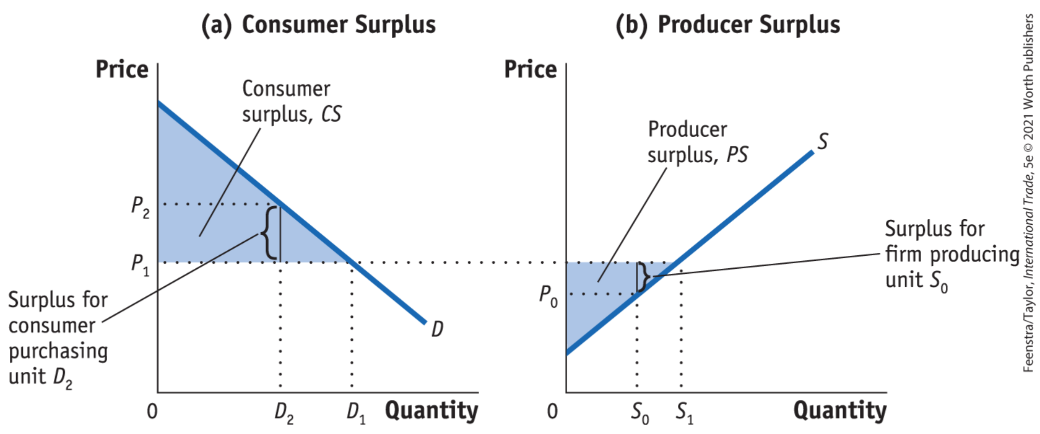
6. The GATT recognizes the ability of blocs of countries to form two types of regional trade agreements:

    i. Free-trade areas, in which a group of countries voluntarily agree to remove trade barriers between themselves
    ii. Customs unions, which are free-trade areas in which the countries also adopt identical tariffs between themselves and the rest of the world

2) The Gains from Trade (part 1)

Consumer and Producer Surplus

FIGURE 1 (1 of 2) Consumer and Producer Surplus

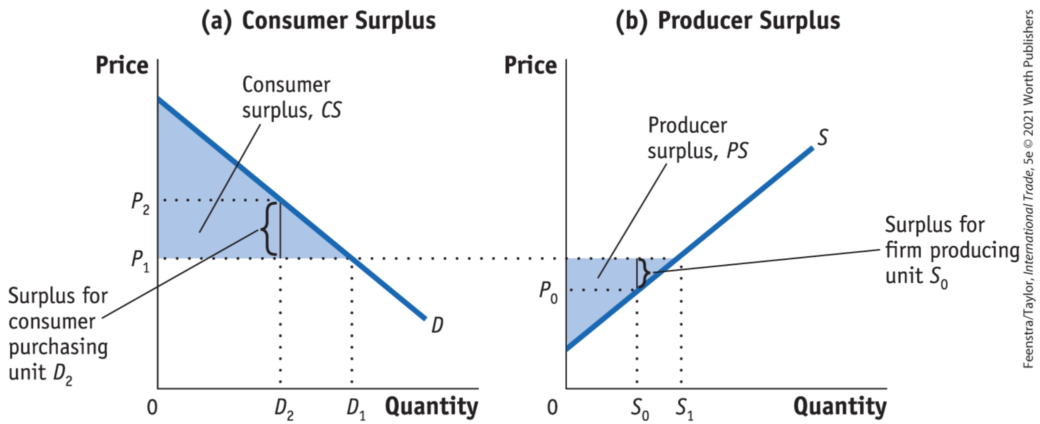


In panel (a), the consumer surplus from purchasing quantity D_1 at price P_1 is the area below the demand curve and above that price. The consumer who purchases D_2 is willing to pay price P_2 but has to pay only P_1. The difference is the consumer surplus and represents the satisfaction of consumers over and above the amount paid.

2) The Gains from Trade (part 2)

Consumer and Producer Surplus

Figure 1 (2 of 2) Consumer and Producer Surplus



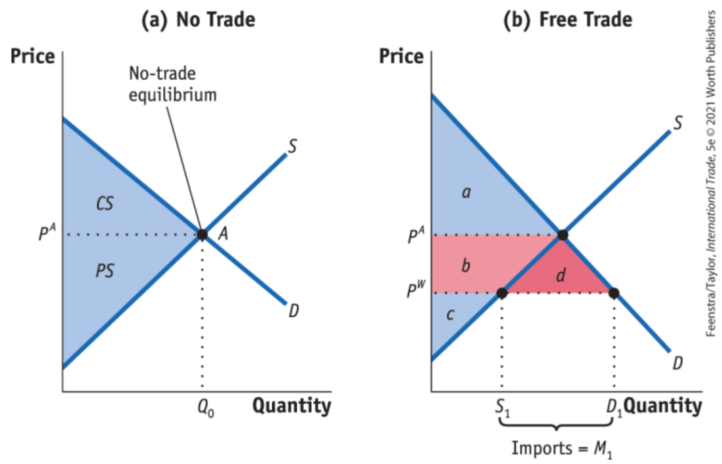
In panel (b), the producer surplus from supplying the quantity S_1 at the price P_1 is the area above the supply curve and below that price. The supplier who supplies unit S_0 has marginal costs of P_0 but sells it for $P_{1.}$ The difference is the producer surplus and represents the return to fixed factors of production in the industry

2) The Gains from Trade (part 3)

Home Welfare

No Trade, Free Trade for a Small Country, Gains from Trade

FIGURE 2



Rise in consumer surplus: +(b + d)
Fall in producer surplus: −b
Net effect on Home welfare: +d

The Gains from Free Trade in Home With Home demand of D and supply of S, the no-trade equilibrium is at point A, at the price P^A producing Q_0.

With trade, the world price is P^W, so quantity demanded increases to D_1 and quantity supplied falls to S_1. Since quantity demanded exceeds quantity supplied, Home imports $D_1 - S_1$.

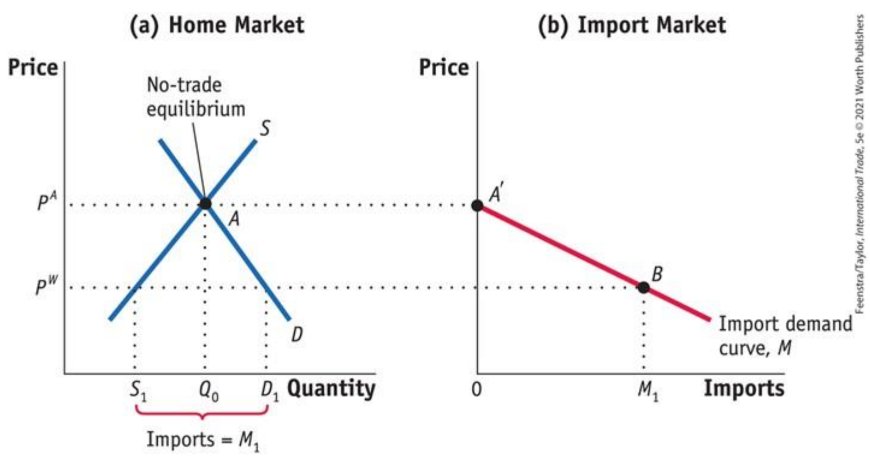
Consumer surplus increases by the area (b + d), and producer surplus falls by area b.

The gains from trade are measured by area d.

2) The Gains from Trade (part 4)

Home Import Demand Curve

FIGURE 3



The import demand curve shows the relationship between the world price of a good and the quantity of imports demanded by Home consumers.

Home Import Demand With Home demand of D and supply of S, the no-trade equilibrium is at point A, with the price P^A and import quantity Q_0.

Import demand at this price is zero, as shown by the point A' in panel (b).

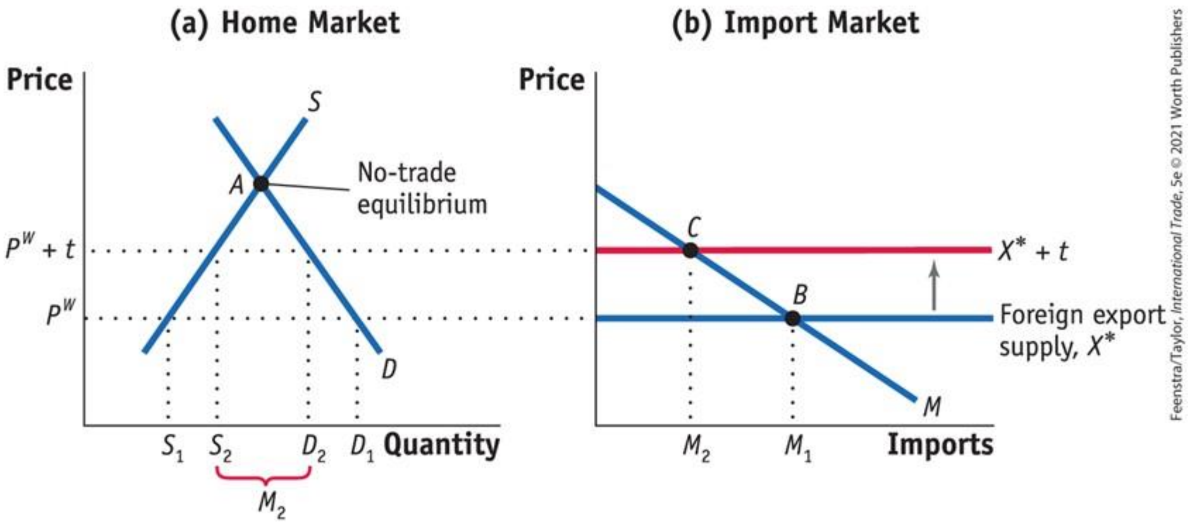
At a lower world price of P^W, import demand is $M_1 = D_1 - S_1$, as shown by point B.

Joining up all points between A' and B, we obtain the import demand curve, M.

3) Import Tariffs for a Small Country (part 1)

Free Trade for a Small Country

FIGURE 4 (1 of 2) Tariff for a Small Country

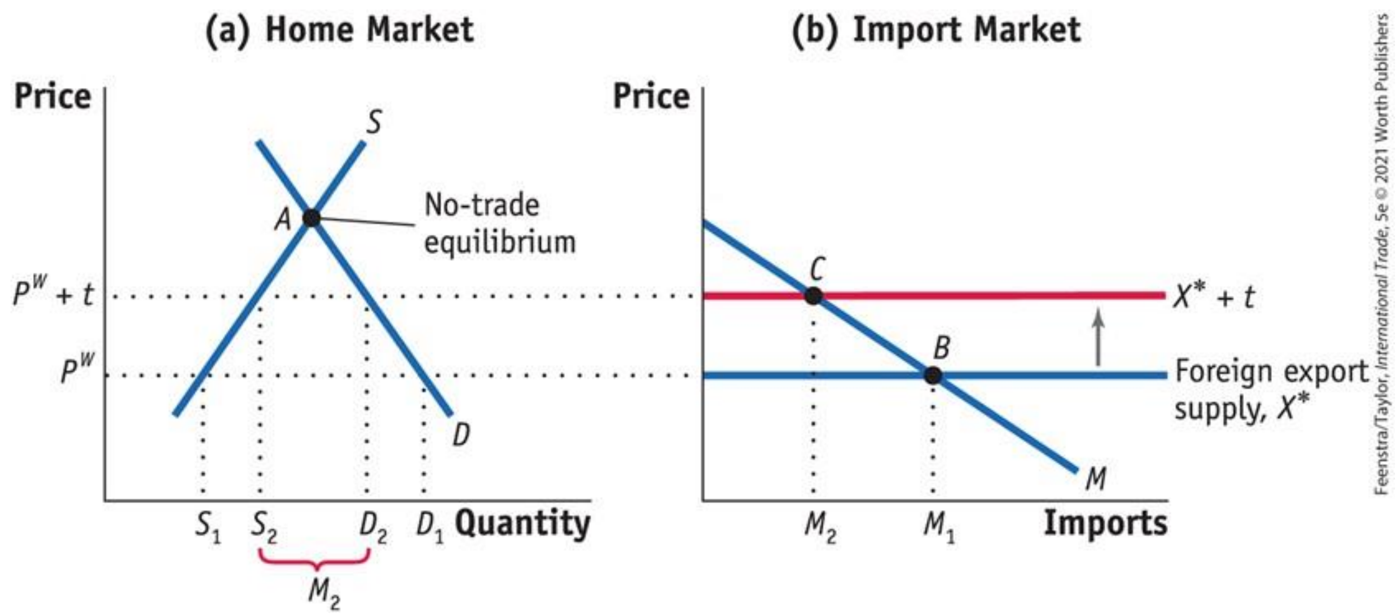


Applying a tariff of t dollars will increase the import price from P^W to $P^W + t$. The domestic price of that good also rises to $P^W + t$. This price rise leads to an increase in Home supply from S_1 to S_2, and a decrease in Home demand from D_1 to D_2, in panel (a).

3) Import Tariffs for a Small Country (part 2)

Free Trade for a Small Country

FIGURE 4 (2 of 2) Tariff for a Small Country

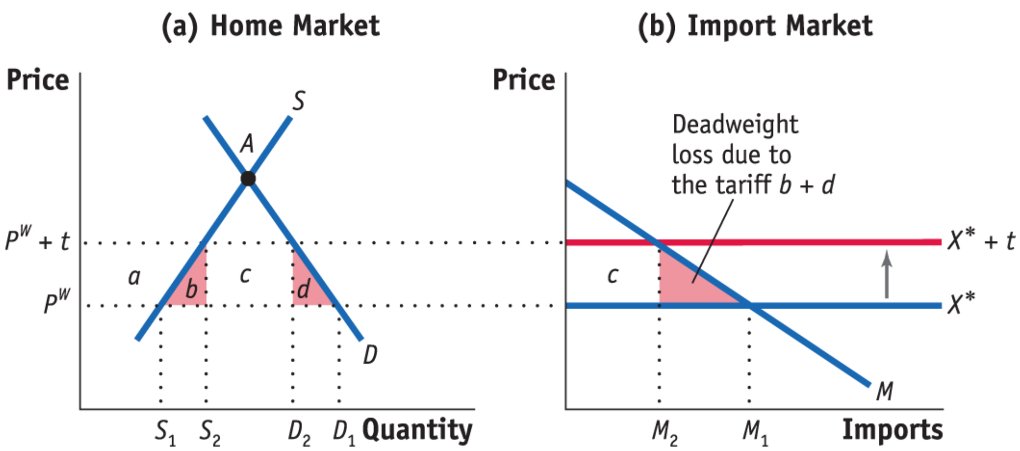


Imports fall due to the tariff, from M_1 to M_2 in panel (b). As a result, the equilibrium shifts from point B to C.

3) Import Tariffs for a Small Country (part 3)

Effect of the Tariff on Consumer Surplus, Producer Surplus, Government Revenue; Overall Effect of the Tariff on Welfare; Production Loss; Consumption Loss

FIGURE 5 (1 of 2)

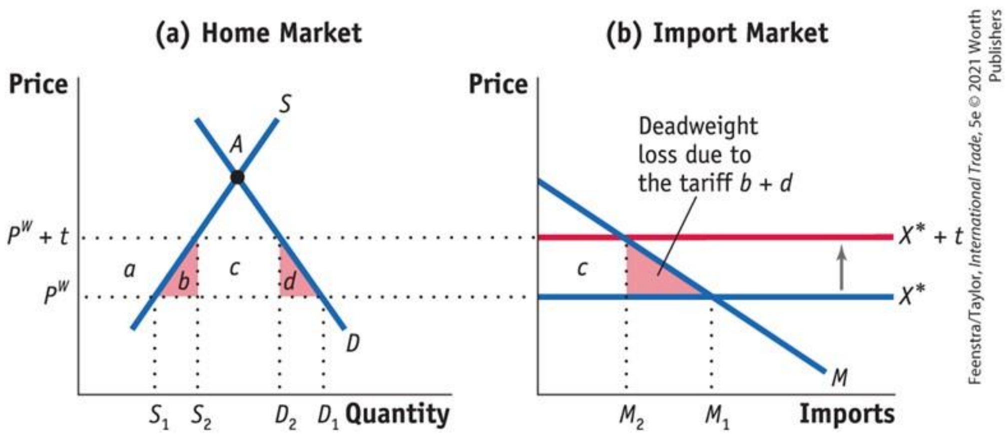


The tariff increases the price from P^W to $P^W + t$. As a result, consumer surplus falls by (a + b + c + d). Producer surplus rises by area a, and government revenue increases by the area c.

3) Import Tariffs for a Small Country (part 4)

Effect of the Tariff on Consumer Surplus, Producer Surplus, Government Revenue; Overall Effect of the Tariff on Welfare; Production Loss; Consumption Loss

FIGURE 5 (2 of 2)



Fall in consumer surplus: −(a + b + c + d)

Rise in producer surplus: +a

Rise in government revenue: +c

Net effect on Home welfare: −(b + d)

Therefore, the net loss in welfare, the deadweight loss to Home, is (b + d), which is measured by the two triangles b and d in panel (a) or the single (combined) triangle b + d in panel (b).

The triangle (b + d) is a deadweight loss, or a loss that is not offset by a gain elsewhere in the economy.

3) Import Tariffs for a Small Country (part 5)

Effect of the Tariff on Consumer Surplus, Producer Surplus, Government Revenue; Overall Effect of the Tariff on Welfare; Production Loss; Consumption Loss

The deadweight loss, triangles (b + d), has the following interpretation:

- The triangle b equals the increase in marginal costs for the extra units produced and can be interpreted as the production loss (or the efficiency loss) due to producing at marginal cost above the world price.

3) Import Tariffs for a Small Country (part 6)

Effect of the Tariff on Consumer Surplus, Producer Surplus, Government Revenue; Overall Effect of the Tariff on Welfare; Production Loss; Consumption Loss

- The triangle d can be interpreted as the drop in consumer surplus for those individuals who are no longer able to consume the units between D1 and D2 because of the higher price. We refer to this drop in consumer surplus as the consumption loss for the economy.

U.S. Tariffs on Steel, 2002-03 (part 1)

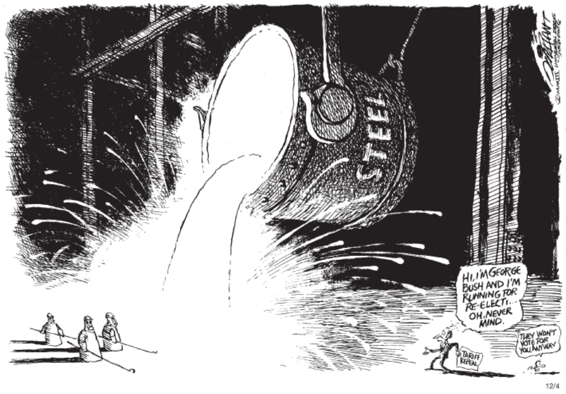
TABLE 1

U.S. ITC Recommended and Actual Tariffs for Steel Shown here are the tariffs recommended by the U.S. International Trade Commission for steel imports in 2002, and the actual tariffs that were applied in 2002.

Product Category	Tariff Recommended by the U.S. ITC (for 2002, %)	Actual U.S. Tariff (used in 2002, %)
Carbon and Alloy Flat Products		
Slab	20	30
Flat products	20	30
Tin mill products	U*	30
Carbon and Alloy Long Products		
Hot-rolled bar	20	30
Cold-finished bar	20	30
Rebar	10	15
Carbon and Alloy Tubular Products		
Tubular products	?**	15
Alloy fittings and flanges	13	13
Stainless and Tool Steel Products		
Stainless steel bar	15	15
Stainless steel rod	?**	15
Stainless steel wire	U*	8

U.S. Tariffs on Steel, 2002-03 (part 2)

Deadweight Loss Due to the Steel Tariff



$$DWL = \frac{1}{2} \cdot \mathrm{t} \cdot \Delta M$$

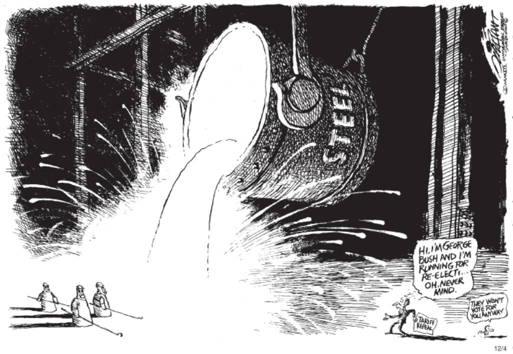
The deadweight loss relative to the value of imports equals:

$$\frac{DWL}{P^W \cdot M} = \frac{1}{2} \cdot \frac{t}{P^W \cdot M} = \frac{1}{2} \cdot \left(\frac{t}{P^W}\right) \cdot \%\Delta M$$

U.S. Tariffs on Steel, 2002-03 (part 3)

Response of the European Countries

- The WTO has a dispute settlement procedure under which countries can bring their complaint and have it evaluated.



- The countries in the European Union (EU) took action by bringing the case to the WTO. The WTO ruling entitled the European Union and other countries to retaliate against the United States by imposing tariffs of their own against U.S. exports.
- The use of tariffs by an importer can easily lead to a response by exporters and a tariff war.

U.S. and Foreign Tariffs Under President Trump, 2018–19 (part 1)

TABLE 2

U.S. and Foreign Tariffs, 2018–19 Panel (a) of this table shows the tariffs applied on U.S. imports by the administration of President Trump during 2018–19. Panel (b) shows the tariffs applied on U.S. exports by foreign countries in retaliation for the U.S. import tariffs.

Product or Country	2017 Tariff (%)	First Date of Tariff*	2018 or 2019 Tariff** (%)	Trade Law
Washing machines	1.3	February 7, 2018	20–50	Section 201
Solar panels	0	February 7, 2018	30	Section 201
Steel	0	March–June, 2018	25	Section 232
Aluminum	2	March–June, 2018	12	Section 232
China (many products)	1.2–3.4	July 2018–October 2019	13.4–26.2	Section 301

Retaliating Country	2017 Tariff (%)	First Date of Tariff*	2018 or 2019** Tariff (%)	In Response to U.S. Tariffs on
China	7.8	April 2018–October 2019	22.7	Steel, aluminum, other products
Mexico	9.6	June 5, 2018	28	Steel, aluminum
Turkey	9.7	June 21, 2018	31.8	Steel, aluminum
European Union	3.9	June 22, 2018	29.2	Steel, aluminum
Canada	2.1	July 1, 2018	20.2	Steel, aluminum
Russia	5.2	August 6, 2018	36.8	Steel, aluminum
India	n.a.	June 15, 2019	10–70	Steel, aluminum

U.S. and Foreign Tariffs Under President Trump, 2018–19 (part 2)

TABLE 8-3

Welfare Impact of 2018 and 2019 U.S. Tariffs Panel (a) shows the welfare impact of the 2018 U.S. tariffs, and the initial 10% tariffs applied to $200 billion of U.S. imports from China in September 2018. Panel (b) extends those results to include the increase in the U.S. tariffs on China during 2019.

	(1)	(2)	(3)
	Consumer Loss from Higher Import Prices	**Tariff Revenue**	**Deadweight Loss**
Annual cost ($ billions)	52.8	36.0	16.8
Annual cost per household ($)	414	282	132

	(1)	(2)	(3)
	Consumer Loss from Higher Import Prices	**Tariff Revenue**	**Deadweight Loss**
Annual cost ($ billions)	106	26.9	79.1
Annual cost per household ($)	831	211	620

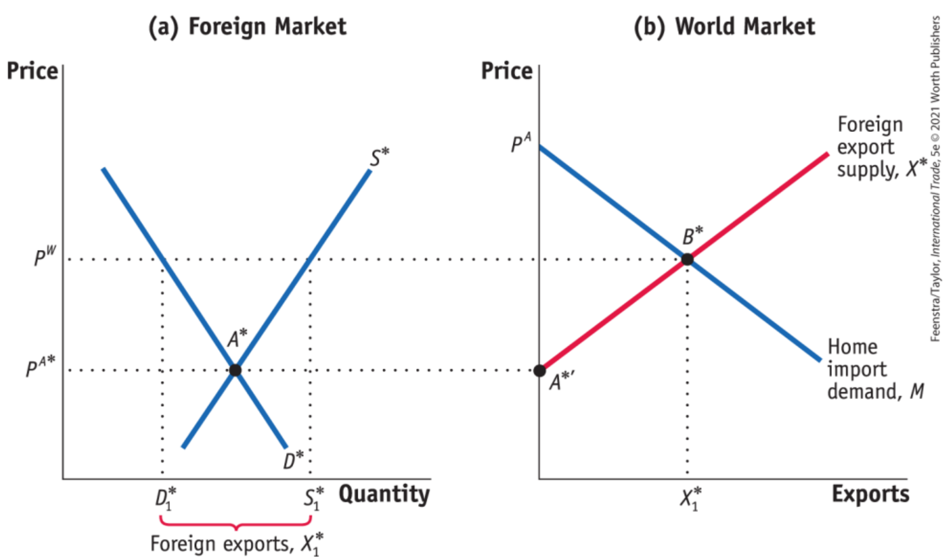
The Political Economy of Tariffs

- If a tariff always leads to a deadweight loss of a small importing country, why then do the United States and many other countries use tariffs as part of their trade policies?
- Politics. The tariff benefits Home producers, so if the government cares more about the producer surplus than consumer surplus, it might decide to use the tariff despite the deadweight loss and consumer costs.
- Because benefits are concentrated and costs are spread more thinly and widely across the country, the industry has the greater political power to persuade the government to use the tariffs.

4) Import Tariffs for a Large Country (part 1)

Foreign Export Supply

FIGURE 6 (1 of 3) Foreign Export Supply



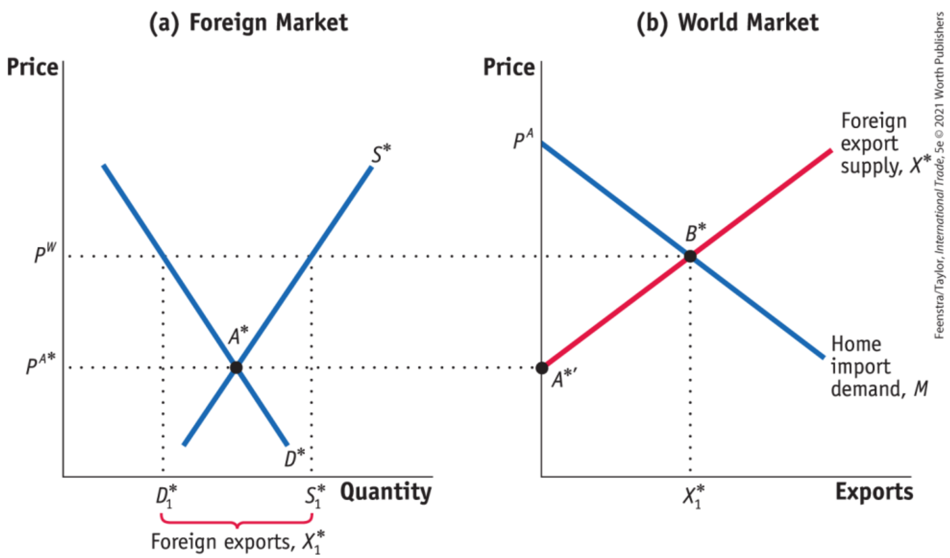
In panel (a), with Foreign demand of D^* and Foreign supply of S^*, the no-trade equilibrium in Foreign is at point A^*, with the price of P^{A*}.

At this price, the Foreign market is in equilibrium and Foreign exports are zero—point A^* in panel (a) and point $A^{*\prime}$ in panel (b), respectively.

4) Import Tariffs for a Large Country (part 2)

Foreign Export Supply

FIGURE 6 (2 of 3) Foreign Export Supply



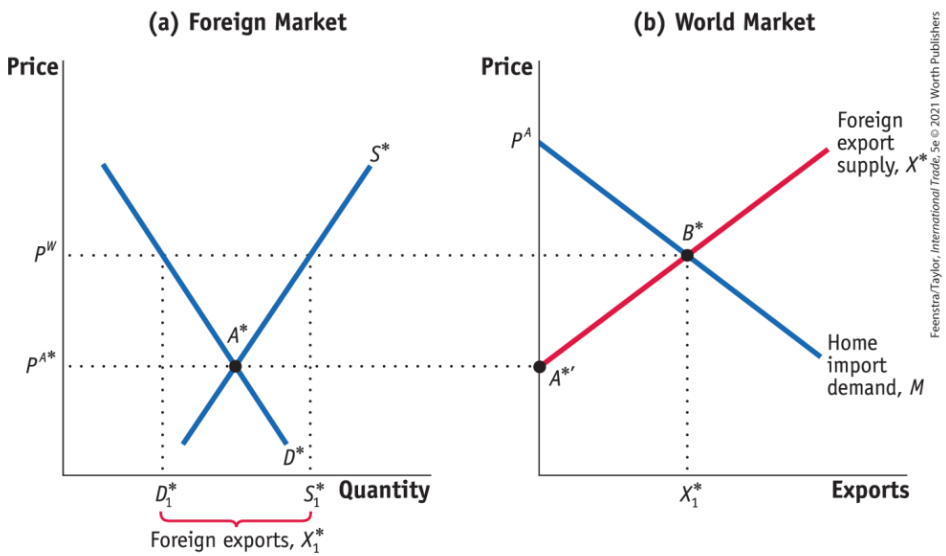
When the world price, P^W, is higher than Foreign's no-trade price, the quantity supplied by Foreign, S^*_1, exceeds the quantity demanded by Foreign, D^*_1, and Foreign exports $X^*_1 = S^*_1 – D^*_1$.

In panel (b), joining up points $A^{*\prime}$ and B^*, we obtain the upward-sloping export supply curve X^*.

4) Import Tariffs for a Large Country (part 3)

Foreign Export Supply

FIGURE 6 (3 of 3) Foreign Export Supply



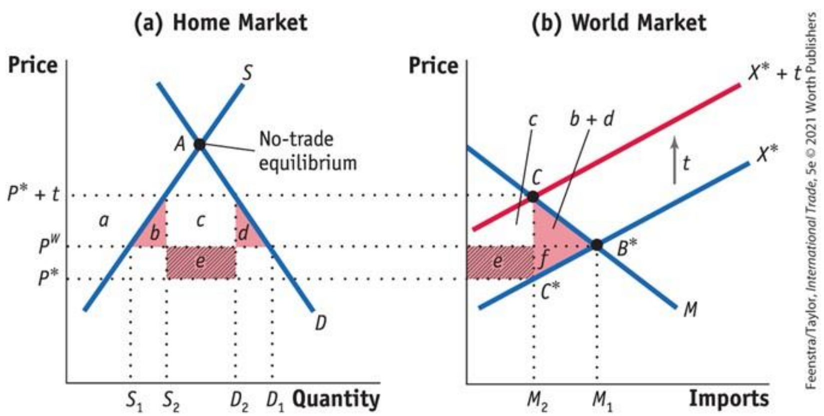
With the Home import demand of M, the world equilibrium is at point B^*, with the price P^W.

4) Import Tariffs for a Large Country (part 4)

Effect of the Tariff

Terms of Trade, Home's Welfare, Foreign and World Welfare

FIGURE 7



Fall in consumer surplus: −(a + b + c + d)

Rise in producer surplus: +a

Rise in government revenue: +(c + e)

Net effect on Home welfare: e − (b + d)

Tariff for a Large Country

The tariff shifts up the export supply curve from X^* to $X^* + t$.

As a result, the Home price increases from P^W to $P^* + t$, and the Foreign price falls from P^W to P^*.

The deadweight loss at Home is the area of the triangle (b + d), and Home also has a terms-of-trade gain of area e.

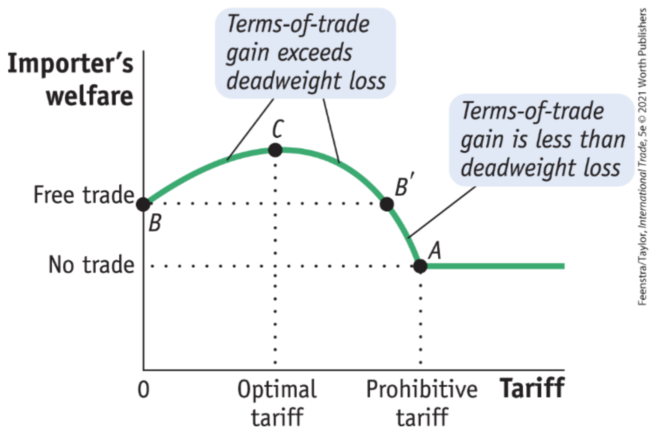
Foreign loses the area (e + f), so the net loss in world welfare is the triangle (b + d + f). Area e is a measure of the terms-of-trade gain for the importer.

4) Import Tariffs for a Large Country (part 5)

Foreign Export Supply

Optimal Tariff for a Large Importing Country

FIGURE 8



Tariffs and Welfare for a Large Country For a large importing country, a tariff initially increases the importer's welfare because the terms-of-trade gain exceeds the deadweight loss. So the importer's welfare rises from point B. Welfare continues to rise until the tariff is at its optimal level (point C). After that, welfare falls. If the tariff is too large (greater than at B′), then welfare will fall below the free-trade level. For a prohibitive tariff, with no imports at all, the importer's welfare will be at the no-trade level, at point A.

Optimal Tariff Formula The formula depends on the elasticity of Foreign export supply, which we call E^*_X:

$$\text{Optimal tariff} = \frac{1}{E^*_X}$$

Conclusions (part 1)

- A tariff on imports is the most commonly used trade policy tool.
- Assuming a small country, when we add together all the effects of a tariff—the drop in consumer surplus, the gain in producer surplus, and the government revenue collected—we still get a net loss for the importing country, which is a deadweight loss resulting from the tariff.
- Why are tariffs used then?
  - They are an easy way for governments to raise revenue.
  - Politics. The government might care more about protecting firms than avoiding losses for consumers.
  - The small-country assumption may not hold in practice.

Conclusions (part 2)

- For a large country, when we add up the drop in consumer surplus, the gain in producer surplus, and the government revenue collected, it is possible for a small tariff to generate welfare gains for the importing country.
- However, in this case, any gain for the importer comes at the expense of the foreign exporters. The drop in the exporter’s welfare due to the tariff is greater than the gain in the importer’s welfare. Therefore, the world loses overall because of the tariff.